



RimeSleepNet: A hybrid deep learning network for s-EEG sleep stage classification

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ABSTRACT

Sleep stage classification is essential for sleep research and clinical diagnostics. However, frequency aliasing in sleep electroencephalogram (s-EEG) signals remains a significant challenge, existing methods have yet to effectively address this issue. This study proposes a hybrid deep-learning model, RimeSleepNet, comprising four key components. First, the rime optimization algorithm adaptively tunes variational mode decomposition (VMD) to reduce frequency aliasing by generating intrinsic mode functions (IMFs). Second, a convolutional neural network (CNN) automatically extracts stage-specific features from IMFs. A multi-head self-attention (MHSA) mechanism then dynamically weights these features to prioritize stage-specific patterns, followed by long short-term memory (LSTM) networks that model temporal dynamics for robust classification of NREM, REM, and WAKE stages. Evaluated on the Chengdu People's Hospital and Sleep-EDF datasets, RimeSleepNet achieves the highest F1 scores of 0.94, 0.89, and 0.92 for NREM, REM, and WAKE stages, respectively, with an AUC of 0.92, outperforming baseline models like CNN and LSTM. Cross-dataset validation confirms its robust generalization (Cohen's $\kappa = 0.90$), and it reduces validation loss by 53 % compared to LSTM, providing an advanced tool for automated sleep stage analysis in sleep disorder diagnosis and personalized monitoring.

1. Introduction

Sleep disorders encompass symptoms that disrupt sleep patterns, duration, and quality. Common manifestations include difficulty initiating sleep, difficulty maintaining sleep, circadian rhythm disruptions, and excessive daytime sleepiness [1,2]. The impact of sleep disorders on individual health is profound and multifaceted. In the short term, poor sleep quality may lead to cognitive impairment, mood instability, memory deficits, and reduced work efficiency [3,4]. Long-term sleep disorders increase the risk of various health issues, including cardiovascular diseases, metabolic syndrome, endocrine disorders, immune dysfunction, and mental health problems [5]. Consequently, early detection and timely intervention are critical for improving the prognosis of individuals with sleep disorders.

s-EEG signals are considered one of the most important predictive markers for sleep disorders [6]. By analyzing s-EEG signals, sleep stages can be identified, sleep quality assessed, and specific abnormal sleep patterns detected, such as sleep apnea and periodic limb movement disorder [7,8]. Traditional s-EEG signal analysis primarily relies on extracting features from the time, frequency, and time-frequency domains, employing classical machine learning algorithms to evaluate sleep states [9,10]. However, these methods often encounter difficulties in capturing nonlinear features, adaptively managing non-stationary signals, efficiently extracting features, and processing high-dimensional amplitude data [11]. Researchers have increasingly adopted deep learning techniques for sleep disorder prediction in response to these challenges. LSTM networks, which can model both long- and short-term dependencies, offer significant advantages in

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analyzing the dynamic features of s-EEG signal time series [12]. The introduction of attention mechanisms has further enhanced this approach. Models that combine LSTM with attention can more effectively focus on the temporal characteristics of s-EEG signals, thereby improving detection accuracy [13]. Zhao et al. proposed a single-channel EEG sleep monitoring model based on bidirectional LSTM, achieving a precision of 84.21 % and an accuracy of 92.73 % [14]. Lee et al. developed an attention-based bidirectional LSTM model for sleep monitoring, which attained a classification accuracy of 85.2 % across wake, REM, and NREM stages [15]. He et al. introduced a bidirectional LSTM-CNN model that integrates bidirectional LSTM with CNN for multi-channel automatic sleep staging, achieving an average classification accuracy of 89.4 % [16]. Zhang et al. proposed four sleep disorder prediction models based on 1D-CNN, Conv-LSTM, 1D-CNN-LSTM, and 2D-CNN-LSTM, with the 1D-CNN-LSTM and 2D-CNN-LSTM models achieving accuracies exceeding 83 %, sensitivities above 81 %, and specificities above 85 % [17]. Additionally, models such as Attention-LSTM [18] and CNN-LSTM [19] have been widely utilized in sleep disorder detection tasks. Nevertheless, these methods still encounter signal decomposition, feature extraction, and optimization limitations.

Due to the distinct characteristics of s-EEG signals in the time-frequency domain, direct analysis may lead to the aliasing of frequency band information. Therefore, decomposing s-EEG signals is a critical step for effective feature extraction. VMD an advanced signal decomposition method, can decompose raw s-EEG signals into multiple IMFs with distinct frequency components. This decomposition enables models to process frequency-specific features, thereby enhancing the detection of sleep disorders [20,21]. Hemant et al. proposed a method based on VMD and linear discriminant analysis that decomposes single-lead ECG signals and extracts relevant features for automatic sleep disorder detection [22]. Mahmud et al. developed a hybrid CNN-bidirectional LSTM model based on VMD that efficiently extracts spectral variation features while maintaining robust individual performance, thus achieving automatic sleep disorder detection [23]. Studies indicate that VMD excels in the spectral analysis of s-EEG signals, revealing abnormal patterns associated with sleep disorders [24]. However, its limitation lies in the empirical selection of decomposition parameters, which poses challenges for parameter optimization and affects decomposition accuracy and stability.

To address these limitations, researchers have employed meta-heuristic algorithms such as Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) to optimize decomposition parameters [25, 26]. These algorithms effectively explore the parameter space, improving decomposition accuracy and facilitating automated parameter tuning. Xiong et al. proposed a GA-optimized VMD algorithm that decomposes s-EEG signals into frequency-specific modal components, significantly enhancing the accuracy of sleep stage classification while preserving signal integrity [27]. However, these methods are less frequently applied in the data preprocessing stage, primarily focusing on subsequent model training. An overreliance on empirical values and traditional methods may lead to local optima, which limits the ability to extract deep information from s-EEG signals [28,29]. Therefore, there is an urgent need for more efficient parameter optimization methods to enhance the accuracy of sleep stage classification.

This paper proposes RimeSleepNet, a hybrid deep learning model. It integrates the rime optimization algorithm to tune VMD parameters adaptively and employs MHSA for dynamic feature weighting, thereby enhancing LSTM-based sleep stage classification and supporting sleep disorder diagnosis. The model introduces three key innovations: (1) A Rime-optimized VMD method is proposed, which automatically determines the number of modes and penalty parameters, reducing signal aliasing and enhancing the accuracy and stability of VMD for s-EEG signal analysis. (2) An MHSA mechanism is introduced to assign dynamic weights to IMFs, highlighting key features related to sleep stages. (3) A hybrid MHSA-LSTM architecture is designed to effectively capture

temporal information and dynamic changes in s-EEG signals, achieving a complete deep learning pipeline from feature decomposition to temporal modeling, thereby improving the accuracy of sleep stage classification. The model is validated on datasets from Chengdu People's Hospital and Sleep-EDF, demonstrating superior performance to traditional and baseline deep learning approaches.

2. Related work

2.1. Signal decomposition addressing frequency aliasing in sleep stage classification

Sleep stage classification relies on extracting frequency-specific features from s-EEG signals to distinguish NREM, REM, and WAKE stages. The frequency components of s-EEG signals, such as delta waves in NREM and alpha waves in REM, reflect the biological characteristics of sleep stages [30]. However, frequency aliasing in s-EEG signals obscures these distinctions, complicating accurate stage identification. VMD addresses this by decomposing s-EEG signals into IMFs, each representing distinct frequency bands, thus reducing aliasing and improving feature extraction efficiency compared to empirical mode decomposition (EMD). Despite its advantages, VMD relies on empirical parameter selection (such as the number of modes and penalty factors). This reliance limits decomposition precision, particularly for nonlinear and non-stationary signals. Optimization techniques, such as PSO and GA, enhance parameter tuning by exploring the parameter space. Nonetheless, these methods are often constrained by local optima and limited preprocessing applications, reducing generalizability across diverse datasets. However, PSO and GA often converge to local optima when configuring VMD for non-stationary s-EEG signals, necessitating adaptive methods, as addressed in this study.

2.2. Dynamic feature weighting and temporal modeling

Effective sleep stage classification requires dynamic weighting of s-EEG signal features to emphasize stage-specific patterns. LSTM networks, widely used for temporal dynamics, lack dynamic weighting mechanisms, limiting their ability to prioritize relevant features [31]. Transformer architectures process time-frequency features but neglect long-term temporal dependencies. Graph convolutional networks (GCNs) model multi-channel spatial relationships but overlook dynamic temporal patterns, reducing generalizability. To address this, dynamic weighting approaches employ MHSA mechanisms to weight IMFs, highlighting stage-specific patterns, and integrate LSTM networks to capture long-term dependencies. These approaches significantly improve classification accuracy and robustness compared to standalone LSTM, Transformer, or GCN models. These methods provide a foundation for RimeSleepNet, which refines MHSA weighting to suppress artifacts and utilizes precise IMFs from configured signal decomposition.

2.3. Positioning of this study

Frequency aliasing in s-EEG signals and the absence of dynamic feature weighting pose significant challenges to accurate sleep stage classification [13,24]. This paper proposes RimeSleepNet, a hybrid deep-learning framework that addresses these challenges through coordinated signal decomposition and dynamic feature weighting. RimeSleepNet adaptively configures VMD parameters using the rime optimization algorithm to minimize aliasing, which avoids local optima through global search strategies and outperforms PSO and GA. It employs MHSA to dynamically weight IMFs, prioritizing stage-specific features while suppressing artifacts, and uses LSTM networks to model temporal dynamics and stage transitions. The coordination of VMD and MHSA improves the extraction of frequency-specific features, enhancing classification performance. RimeSleepNet demonstrates improved classification accuracy and robustness compared with PSO, GA,

Transformer, and GCN methods. These improvements are validated on datasets from Chengdu People's Hospital and the Sleep-EDF database, providing a practical approach for sleep stage classification and disorder diagnosis.

3. Methodology

3.1. Problem formulation

s-EEG signal frequency aliasing and feature differences across stages hinder accurate sleep stage classification. The task is defined as: given a multi-channel s-EEG signal sequence $S = \{S_n\}_{n=1}^N$, where $S = \{X_c^{(n)}\}_{c=1}^C$ represents a 30-s segment across C channels, each segment is classified as $y^{(n)} \in \{NREM, REM, WAKE\}$. Sleep stage classification identifies abnormal patterns, providing a basis for diagnosing sleep disorders such as insomnia or sleep apnea. This study addresses these challenges by adaptively configuring VMD parameters to reduce aliasing and dynamically weighting IMFs to emphasize stage-specific features, forming the basis for the RimeSleepNet framework.

3.2. Model overview

The Rime-optimized sleep stage classification network (RimeSleepNet) model architecture, shown in Fig. 1, integrates four submodules to optimize feature extraction and classification performance. As illustrated in Fig. 2, the RimeSleepNet model is developed to achieve high-precision sleep stage classification (NREM, REM, WAKE) from s-EEG signals. The model comprises four submodules: the rime optimization algorithm is used to tune VMD parameters adaptively; VMD decomposes signals into IMFs; an MHSA mechanism dynamically weights IMFs to prioritize stage-specific features; and LSTM networks model temporal dynamics for stage classification. Classification results are generated through a fully connected layer, providing a basis for diagnosing sleep disorders.

3.3. Signal decomposition and optimization

3.3.1. Rime optimization

The rime optimization algorithm adaptively tunes the VMD [32,33] parameters, specifically the mode number K and penalty factor α , as illustrated in Fig. 2(a), to overcome the local optima limitations of traditional algorithms such as PSO and GA. For s-EEG signals $X_{EEG}^c(t)$ (channel $c \in [1, 8]$), a frost particle population $R_c = \{[k_{i1}, \alpha_{i1}], \dots, [k_{iL}, \alpha_{iL}]\}$ is initialized, with $K \in [4, 12]$, $\alpha \in [10, 4000]$, ranges set based on s-EEG frequency characteristics. Particle count is set to 50, maximum iterations $T = 20$, and convergence threshold 10^{-4} , weighting factor $\lambda = 0.1$, and step size parameter $\omega = 5$, as detailed in Table 1. Particle positions are updated as:

$$p_i^{(n+1)} = R_b + c_1 \cdot \cos(\theta) \cdot \alpha(h \cdot (U - L) + L), c_2 < E. \quad (1)$$

Where R_b represents the global optimum, $c_1 \in (-1, 1)$ is a random factor, $\theta = \frac{\pi \cdot t}{10 \cdot T}$ is a direction control factor, $h \in (0, 1)$ denotes adhesion strength, $E = \frac{t}{T}$ is a convergence criterion, and t and T are the current and maximum iterations, respectively. The step size is adjusted by $\beta =$

$1 - \frac{|\omega - \frac{\omega}{T}|}{\omega}$, with $\omega = 5$. The objective function is defined as:

$$L_c(k, \alpha) = \frac{1}{N} \sum_{n=1}^N \left\| X_{EEG}^c(t) - \sum_{i=1}^k IMF_{c,i}(t) \right\|_2^2 + \lambda \sum_{i=1}^k \sigma(IMF_{c,i}(t)). \quad (2)$$

Where $\sigma(IMF_{c,i}(t))$ represents the modal spectral width, and $\lambda = 0.1$ balances reconstruction error and spectral separation. Optimization converges after $T = 20$ iterations or when the loss change falls below 10^{-4} , yielding optimal parameters $[k_c^*, \alpha_c^*]$. Compared to PSO and GA, the rime optimization algorithm enhances global search capabilities through dynamic direction control and adhesion mechanisms, significantly reducing the risk of local optima. Parameters are detailed in Table 1.

3.3.2. VMD analysis

VMD is applied to decompose s-EEG signals into K IMFs, as illustrated in Fig. 2(b), optimizing modal center frequencies ω_i and spectral widths to extract frequency components. The optimization objective is:

$$L_{VMD} = \sum_{i=1}^k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_i(t) \cdot e^{-j\omega_i t} \right] \right\|_2^2 + \alpha \left\| X_{EEG}(t) - \sum_{i=1}^k u_i(t) \right\|_2^2. \quad (3)$$

In each iteration, the i -th modal function $u_i(t)$ is updated as:

$$u_i^{(n+1)}(t) = \frac{X_{EEG}(t) - \sum_{j \neq i} u_j^{(n)}(t) + \lambda \cdot \left(\hat{X}_{EEG}(t) - \hat{u}_i^{(n)}(t) \right)}{1 + 2\alpha}. \quad (4)$$

The center frequencies ω_i are updated as:

$$\omega_i^{(n+1)} = \frac{\int_{-\infty}^{\infty} t \cdot |u_i(t)|^2 dt}{\int_{-\infty}^{\infty} |u_i(t)|^2 dt}. \quad (5)$$

Global reconstruction error is corrected via the Lagrange multiplier λ :

$$\lambda^{(n+1)} = \lambda^{(n)} + \tau \left(X_{EEG}(t) - \sum_{i=1}^k u_i(t) \right). \quad (6)$$

With an update step size $\tau = 0.1$. The decomposition yields $X_{EEG}(t) = \sum_{i=1}^k u_i(t)$, with IMFs corresponding to EEG frequency bands ($\delta: 0.5 \sim 4$ Hz, $\theta: 4 \sim 8$ Hz, $\alpha: 8 \sim 12$ Hz; $\beta: 12 \sim 30$ Hz), effectively distinguishing overlapping frequency characteristics of WAKE, REM, and NREM through spectral width optimization.

3.4. Sleep stage classification

3.4.1. Feature extraction and selection

IMFs obtained from VMD are processed through a CNN to automatically extract features capturing stage-specific characteristics of s-EEG signals. As illustrated in Fig. 2(c), a multi-layer 1D CNN, consisting of two convolutional layers with 64 and 128 filters (kernel size of 5), followed by max-pooling layers (pool size of 2), extracts local frequency and temporal patterns from the IMFs. The input IMFs, shaped as (N, T, C) (N : number of samples, $T = 6000$, 30-s at 200 Hz, C = number of IMFs), are transformed into feature maps of shape (N, T', F) , where $T' = 1500$

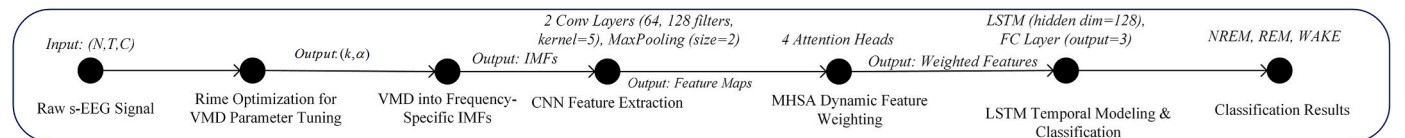


Fig. 1. RimeSleepNet: Model architecture for s-EEG based sleep stage classification.

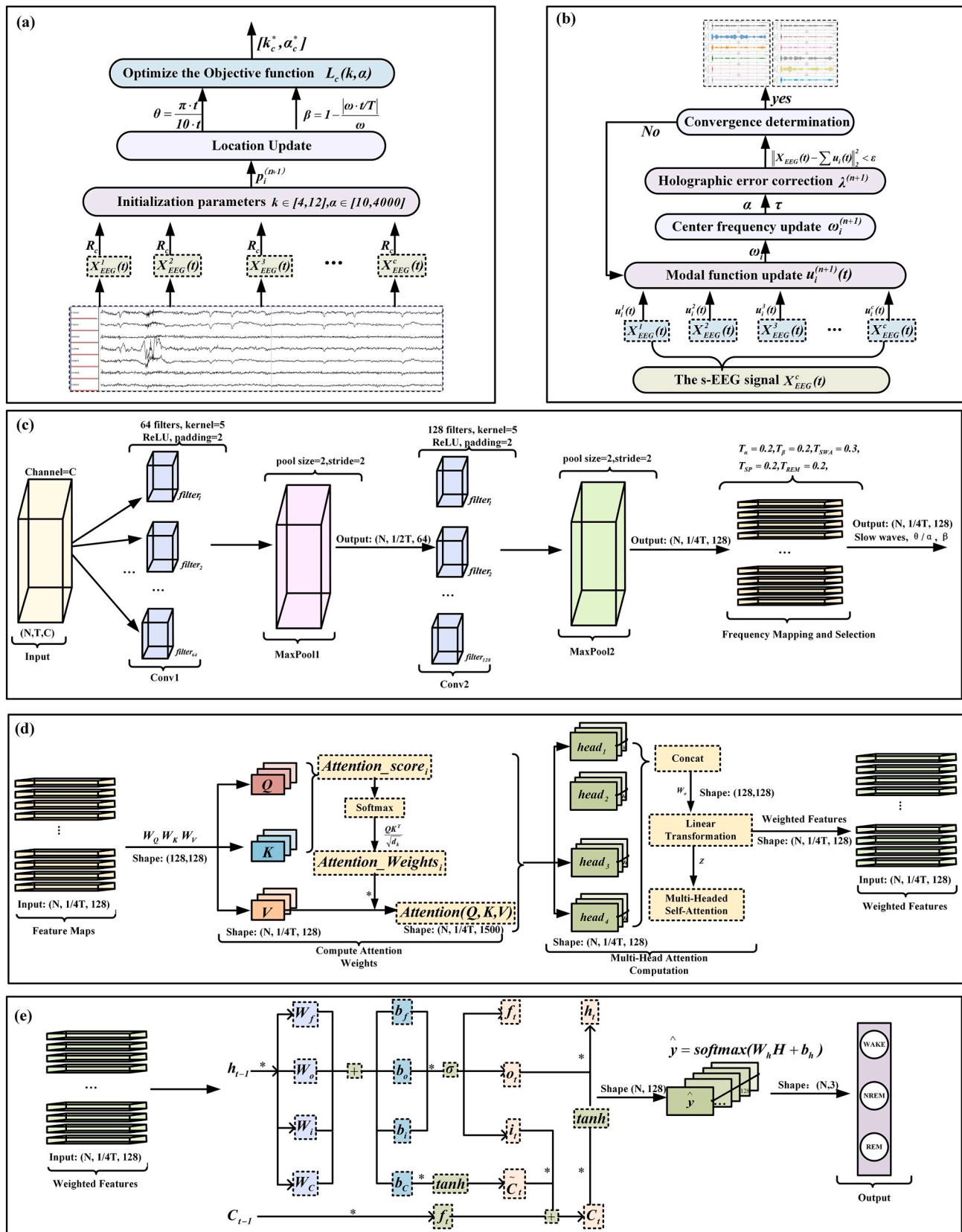


Fig. 2. RimeSleepNet model architecture for sleep stage classification. (a) Rime optimization for adaptive VMD parameter tuning. (b) VMD into frequency-specific IMFs. (c) CNN-based automated extraction of sleep stage features. (d) MHSA-driven dynamic weighting of features. (e) LSTM-based temporal modeling for stage classification.

Table 1
RIME optimization parameters and convergence.

Parameter	Value	Description
Particle Count	50	Size of the optimization population
Maximum Iterations	20	Maximum number of optimization iterations
Convergence Threshold	10^{-4}	Stopping condition for loss change
Weighting Factor	0.1	Balances reconstruction error and spectral separation
Step Size Parameter	5	Adjusts optimization step size
Mode Number Range	[4~12]	Search range for VMD mode num
Penalty Factor Range	[10,4000]	Search range for VMD penalty factor
Convergence Iterations	[15~20]	Iterations required for objective function stabilization

(after pooling) and $F = 128$ (feature dimension).

To align with stage-specific frequency characteristics, CNN-extracted features are mapped to specific frequency bands via Fourier transform, including slow waves (0.5–2 Hz), θ/α waves (4–13 Hz), and β waves (8–30 Hz), generating frequency band energy features. The frequency band energy E_b for a given band is computed as:

$$E_b = \sum_{f=f_{low}}^{f_{high}} |X(f)|^2. \quad (7)$$

Where $X(f)$ represents the Fourier transform of the CNN feature map, and f_{high} and f_{low} denote the lower and upper frequency bounds of band. The dominant frequency f_{dom} within each band is determined as:

$$f_{dom} = \arg \max_{f \in [f_{low}, f_{high}]} |X(f)|^2. \quad (8)$$

Where $|X(f)|^2$ is the power spectral density (PSD) at frequency f .

Feature selection adheres to the standards of the American Academy of Sleep Medicine (AASM) [34]. The energy ratios of frequency bands (e.g., slow waves, θ/α , β) are computed from the PSD of CNN-extracted features and normalized to determine selection weights. These weights are then combined with predefined parameters $T_\alpha = 0.2$, $T_\beta = 0.2$, $T_{SWA} = 0.3$, $T_{SP} = 0.2$, and $T_{REM} = 0.4$ as detailed in Table 2, to select feature dimensions most relevant to each sleep stage. For instance, feature dimensions with higher energy in the slow wave band (0.5–2 Hz) are prioritized for NREM, while those in the θ/α band (4–13 Hz) are prioritized for REM. The selected feature dimensions, weighted by these parameters, provide a preliminary prioritization of stage-specific characteristics, which is further refined through dynamic weighting in Section 3.4.2. This ensures differentiation of WAKE (β waves, 12–30 Hz), REM (θ/α waves, 4–13 Hz), and NREM (slow waves, 0.5–2 Hz) features.

3.4.2. MHSA-LSTM classification

The feature maps extracted by the CNN, with a shape of (N, T', F) , are dynamically weighted using an MHSA mechanism, as illustrated in Fig. 2(d), to highlight stage-specific patterns in s-EEG signals identified

Table 2
Feature selection parameters and physiological basis.

Parameter	Value	Frequency Band	Sleep Stage	Physiological Basis
$T_\alpha = 0.2$	0.2	α waves (8–12 Hz)	WAKE, REM	α waves are prominent in wakefulness and REM
$T_\beta = 0.2$	0.2	β waves (12–30 Hz)	WAKE	β waves reflect high-frequency activity in wakefulness
$T_{SWA} = 0.3$	0.3	Slow waves (0.5–2 Hz)	NREM	Slow waves dominate deep NREM sleep
$T_{SP} = 0.2$	0.2	Spindles (12–14 Hz)	NREM (N2)	Spindles are characteristic of N2 stage
$T_{REM} = 0.4$	0.4	θ/α waves (4–13 Hz)	REM	θ/α waves predominate in REM

through preliminary weighting. The preliminary weighting enhances the relevance of features associated with distinct frequency bands. The MHSA module, equipped with four attention heads, assigns weights based on contextual relevance. Specifically, MHSA generates query, key, and value matrices:

$$Q = XW_Q, K = XW_K, V = XW_V. \quad (9)$$

Where X is the input feature map, and W_Q , W_K , and W_V are learnable weight matrices. Attention weights are computed as:

$$Attention_Weights = \text{soft max} \left(\frac{QK^T}{\sqrt{d_k}} \right). \quad (10)$$

$$Attention(Q, K, V) = Attention_Weights \cdot V. \quad (11)$$

Where $d_k = 32$ is the dimension of each attention head (F/h , with $h = 4$, number of attention heads). The attention weights, shaped as (N, T', T') , reflect the similarity between time steps, dynamically adapting to temporal context. For instance, time steps with prominent slow wave features receive higher weights during NREM stages. In contrast, REM stages prioritize time steps with θ/α wave patterns, enabling the model to focus on stage-specific characteristics across the sequence. The multi-head output is generated as follows:

$$MHSA(Q, K, V) = \text{Concat}(head_1, head_2, \dots, head_h)W_O. \quad (12)$$

Where W_O is a learnable output matrix, producing weighted features of shape (N, T', F) .

The weighted features are then processed by LSTM networks, as illustrated in Fig. 2(e), to model the temporal dynamics of s-EEG signals. The LSTM updates its hidden state across the sequence of time steps as follows:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f), i_t = \sigma(W_i[h_{t-1}, z_t] + b_i). \quad (13)$$

$$\tilde{C}_t = \tan h(W_C[h_{t-1}, z_t] + b_C), C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t. \quad (14)$$

$$o_t = \sigma(W_o[h_{t-1}, z_t] + b_o), h_t = o_t \cdot \tan h(C_t). \quad (15)$$

The final hidden state h_T (dimension 128) encapsulates the temporal dynamics of the sequence. Classification outputs are generated through a fully connected layer:

$$\hat{y} = \text{soft max}(W_h h_T + b_h). \quad (16)$$

Where $\hat{y}$, shaped as $(N, 3)$, represents the probabilities for WAKE, NREM, and REM stages. Compared to Transformer models, the gating mechanisms of LSTM (the forget and input gates) more effectively model long-term dependencies in s-EEG signals, rendering it suitable for dynamic stage transitions. In contrast, the global attention mechanism of Transformers may allocate suboptimal weights due to artifact interference. However, the LSTM's dynamic adjustment of feature relevance enhances classification robustness. The four submodules of RimeSleepNet operate synergistically. The Rime optimization ensures that the VMD parameters adapt to individual variability. The VMD provides artifact-resistant IMFs. The MHSA weighting prioritizes stage-specific features, while LSTM modeling captures temporal dynamics. Together, these components achieve high-precision classification of sleep stages.

4. Experimental setup and results

4.1. Datasets and preprocessing

To address artifact interference and cross-dataset individual variability, the performance of RimeSleepNet sleep stage classification is evaluated on a self-collected dataset from Chengdu People's Hospital and the public Sleep-EDF dataset [35,36], with detailed information provided in Table 3. This work is based on a prospective study in which

Table 3
Dataset overview.

Dataset	Subjects	Channels	Sampling Rate	W	N1	N2	N3	REM	Total Samples (30 s)
Hospital	50	C3-M2, C4-M1, F3-M2, F4-M1, O1-M2, O2-M1	200Hz	10413 (17.71 %)	6780 (11.53 %)	23609 (40.15 %)	9690 (16.48 %)	8308 (14.13 %)	58800
Sleep-EDF [35,36]	78	Fpz-Cz, Pz-Oz	100Hz	65951 (33.7 %)	21522 (11.0 %)	69132 (35.4 %)	13039 (6.7 %)	25835 (13.2 %)	195479

Note. Hospital dataset: n = 50, age 18–65 years; male n = 30, female n = 20. Sleep-EDF dataset (sleep-cassette subset, healthy adults): n = 78, age 25–101 years; male n = 44, female n = 34. Across both datasets: n = 128 subjects and 254,279 30-s epochs ($\approx$ 2,119 h). Clarification: we used the sleep-cassette subset only; the sleep-telemetry subset (participants with mild difficulty falling asleep; n = 22) was not included.

the Chengdu People's Hospital dataset was collected by recruiting healthy volunteers and patients with sleep disorders under predefined inclusion and exclusion criteria. In addition, the publicly available Sleep-EDF dataset, representing retrospective data, was also included for model evaluation. Participants in the Chengdu People's Hospital dataset were prospectively recruited according to explicit inclusion and exclusion criteria. Inclusion criteria were: (1) age between 18 and 65 years; (2) either healthy volunteers without self-reported sleep disorders or patients with clinically diagnosed common sleep disorders, such as insomnia or sleep apnea, defined by the International Classification of Sleep Disorders (ICSD-3); (3) the ability to undergo overnight s-EEG monitoring and complete all study procedures. Exclusion criteria were: (1) the presence of severe psychiatric disorders (such as schizophrenia or major depressive disorder) or neurological conditions (such as epilepsy or Parkinson's disease) that could potentially influence EEG patterns; (2) the current use of medications known to significantly alter sleep architecture within two weeks prior to data acquisition; (3) EEG recordings of insufficient quality due to excessive artifacts or missing segments. All study procedures were reviewed and approved by the Ethics Committee of Chengdu People's Hospital, and written informed consent was obtained from all participants prior to data collection. Data preprocessing, which involves wavelet denoising and bandpass filtering to eliminate artifacts while retaining physiological bands, is illustrated in Fig. 3.

4.2. Training configuration and evaluation metrics

The training configuration of RimeSleepNet in Table 4 is designed to optimize classification performance for WAKE, NREM, and REM sleep stages while ensuring cross-dataset robustness. The model is trained on an NVIDIA RTX 3090 GPU using the PyTorch framework, with a batch size of 48, a learning rate of 1×10^{-4} , and the Adam optimizer. Training spans 40 epochs and uses a weighted cross-entropy loss function with class weights (NREM: 0.8, REM: 1.3, WAKE: 0.8) to balance class

Table 4
RimeSleepNet model configuration parameters.

Parameter Category	Parameter Name	Set Value
Network Structure	Input Dimension	(N, T, C)
	Hidden Layer Dimension	128
	Number of Attention Heads	4
	Dropout Rate	0.3
Training Parameters	Batch Size	48
	Learning Rate	1e-4
	Weight Decay	1e-5
	Training Epochs	40
Class Weights	NREM	0.8
	REM	1.3
	WAKE	0.8

distribution and improve REM classification accuracy by addressing frequency overlap. Five-fold cross-validation is applied, with 80 % of the data for training and 20 % for validation, to ensure classification consistency across subjects and datasets.

All statistical analyses were performed using SPSS version 26.0 and Python (SciPy package). Model performance metrics, including accuracy, precision, recall, F1-score, and AUC, were reported as mean $\pm$ standard deviation (SD) across five-fold cross-validation. Differences between RimeSleepNet and baseline models (CNN, LSTM, VMD-CNN, VMD-LSTM, RIME-VMD-CNN, RIME-VMD-LSTM) were evaluated using two-tailed paired t-tests for pairwise comparisons. When normality assumptions were violated, as assessed by the Shapiro-Wilk test, the Wilcoxon signed-rank tests were applied. Statistical significance was considered at $p < 0.05$, and Bonferroni correction was applied for multiple comparisons.

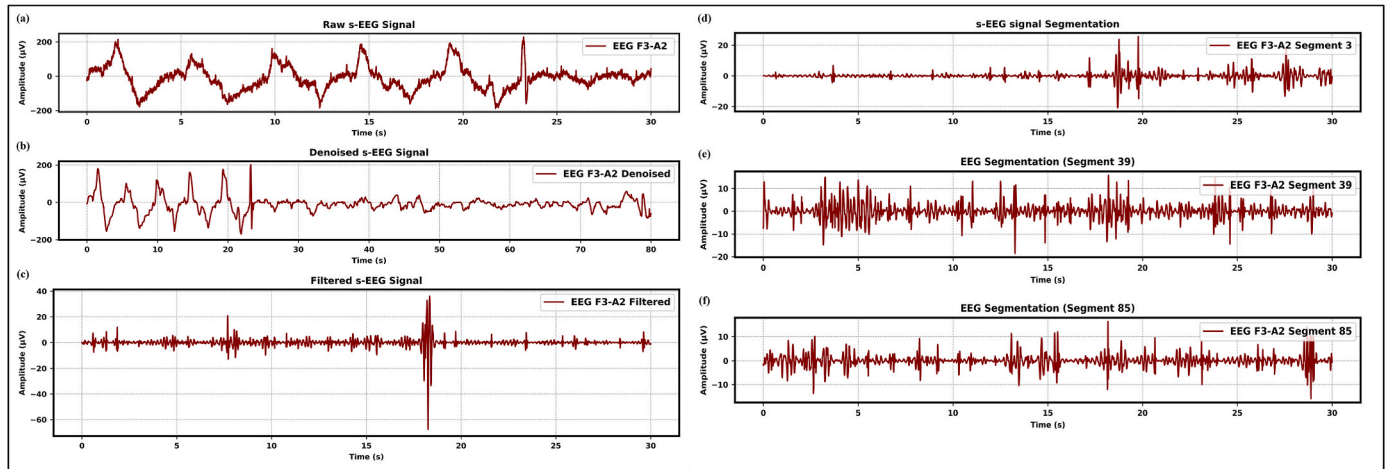


Fig. 3. s-EEG signal processing stages showing dynamic changes across segments. (a) Raw EEG signal. (b) Denoised signal. (c) Filtered signal. (d–f) Segments 3, 39, and 85.

4.3. RIME optimization and VMD results

4.3.1. RIME optimization results

(1) Single-Channel Optimization

The optimization of VMD parameters is accomplished through the RIME algorithm, which automatically adjusts the mode number K and the penalty factor α , thereby overcoming the limitations associated with empirical parameter selection. The optimization process for the single-channel F3-A2 is detailed in Table 5, where the initial parameters $K = 5$, $\alpha = 2000$ (loss = 6.75) converge to $K = 9$, $\alpha = 1815$ (loss = 5.95) after 20 iterations, resulting in an 11.85 % reduction in loss.

As shown in Table 5, the initial parameters ($K = 5$, $\alpha = 2000$) yield a loss of 6.75, indicating a substantial decomposition error from empirical settings. In the second iteration ($K = 4$, $\alpha = 2873$), the loss increases to 7.71 (an increase of 14.2 %), suggesting that a low mode number and a high penalty factor reduce decomposition accuracy. The third iteration ($K = 4$, $\alpha = 3925$) further elevates the loss to 8.01, confirming that increasing α alone is detrimental. In the fourth iteration ($K = 6$, $\alpha = 2451$), the loss decreases to 6.68 (a decrease of 1.03 %), indicating improved accuracy. The eighth iteration ($K = 11$, $\alpha = 1679$) reduces the loss to 6.05 (a decrease of 10.37 %), demonstrating enhanced decomposition with a higher mode number and a lower penalty factor. By the fifteenth iteration ($K = 9$, $\alpha = 1815$), the loss reaches a minimum of 5.95 (a decrease of 11.85 %), achieving optimal parameters that provide high-quality IMFs for subsequent classification.

(2) Multi-Channel Optimization

The core objective of multi-channel optimization is to independently optimize K and α for each channel to adapt to the frequency characteristics and signal complexity of different channels. The RIME optimization results are shown in Table 6, the optimal K is concentrated in 9, 10, and 11, indicating that these parameters can be well adapted to the frequency distribution. The α range of different channels is from 1254–3245, indicating that the requirements of different signals for frequency concentration vary greatly.

Fig. 4 verifies the effectiveness of the Rime algorithm in dynamically searching for the number of modes and the penalty parameters. With the increase in the number of iterations, the collaborative optimization of the mode number K and penalty factor α continuously decreases the value of the objective function. Finally, it obtains the optimal parameter combination. This result verifies the effectiveness of the two-parameter adaptive adjustment mechanism. It also provides parameter support for the subsequent accurate decomposition of the s-EEG signal.

4.3.2. VMD results

As shown in Fig. 5, the IMFs obtained from VMD of the s-EEG signal are decomposed into distinct frequency components.

Low-frequency modes (IMF1~IMF3): IMF1 is identified as the mode

Table 5
RIME optimization parameters (Single channel F3-A2).

Iteration	K	α	Loss	Iteration	K	α	Loss
1	5	2000	6.75	11	11	1180	6.07
2	4	2873	7.71	12	7	1976	6.51
3	4	3925	8.01	13	10	3549	6.20
4	6	2451	6.68	14	10	3059	6.03
5	5	1549	6.64	15	9	1815	5.95
6	7	3614	6.70	16	8	2056	6.09
7	8	3863	6.50	17	8	4342	6.54
8	11	1679	6.05	18	8	1609	6.35
9	7	2276	6.54	19	11	4682	6.17
10	6	3198	6.80	20	6	1841	6.59
Optimal parameters		$K = 9$	$\alpha = 1815$			Loss = 5.95	

Table 6

Optimized parameters of multiple channels for rime.

Channel	Iteration	K	α	Loss
F3-A2	15	10	3197	1.27
F4-A1	6	11	1279	0.02
A1-A2	15	9	3245	0.06
C3-A2	6	10	1422	1.44
C4-A1	5	9	1659	1.24
O1-A2	11	11	1313	0.95
O2-A1	19	11	1254	0.14

with the highest energy and lowest frequency, capturing low-frequency fluctuations in the s-EEG signal. IMF2 is recognized as the second-lowest-frequency mode, with a slightly higher frequency than IMF1 and a lower amplitude than IMF1. IMF3 has a further increased frequency and captures slow-wave components, particularly during the NREM stage.

Intermediate frequency modes (IMF4~IMF7): The amplitude of IMF4 is reduced while its frequency is increased, potentially being associated with θ wave activity, and is typically observed in the WAKE and REM stages. The amplitudes of IMF5 and IMF6 are further reduced, yet changes in the intermediate frequency (θ to α wave) region are still captured, particularly during the REM stage. The frequency of IMF7 is further increased, with mid-frequency characteristics of the REM stage being reflected.

High-frequency modes (IMF8~IMF9): The amplitudes of these modes are significantly reduced, and high-frequency artifacts or short-term activities are primarily captured, with higher levels of high-frequency activity typically observed during the REM stage. The frequency of IMF8 is further elevated, and short-term high-frequency activities are captured, particularly during the REM stage. The waveform of IMF9 resembles random noise and is associated with electrode contact noise and motion artifacts; pronounced high-frequency vibrations are observed during the REM stage.

Fig. 6(a) presents the IMFs obtained from VMD ($K = 5$, $\alpha = 2000$), with each IMF corresponding to a distinct frequency component. As detailed in Table 7, low-frequency modes (IMF1~3, 0.5–7 Hz) account for 54.60 % of the total energy, with IMF1 exhibiting the highest energy ($15434.2\mu V^2$, 37.20 %) and the lowest relative correlation (25.13 %), underscoring the dominance and independence of low-frequency components in the s-EEG signal. Relative correlations of IMF2 and IMF3 are 48.27 % and 33.94 %, respectively, indicating some overlap. However, the energy distribution of IMF4 and IMF5 (17.08 % and 28.33 %) and their high correlations (49.39 % and 43.27 %) suggest a concentration that may lead to mode mixing and loss of mid-to-high-frequency information.

Fig. 6(b) and (c) depict the IMFs derived from GA-VMD and PSO-VMD, respectively. Table 8 shows that the low-frequency modes (IMF1~3) contribute only 37.12 % to the total energy, significantly lower than VMD's 54.60 %. Moreover, IMF6 exhibits an excessively high energy ($14025.34\mu V^2$, 33.97 %), indicating that GA-VMD prioritizes mid-to-high-frequency components while neglecting the importance of low-frequency modes, with its relative correlations fluctuating (20.19 %–41.22 %), suggesting unstable mode independence. Table 9 reveals that the low-frequency modes account for 39.48 %, still below VMD, with IMF4 showing an extremely low energy ($421.58\mu V^2$, 1.02 %), potentially leading to the loss of critical information. Additionally, the relative correlations in PSO-VMD fluctuate widely (13.18 %–36.30 %), reflecting instability in the decomposition process.

In contrast, Table 10 highlights the superiority of the RIME-VMD method. Low-frequency modes (IMF1~3) contribute 28.54 % to the total energy, lower than VMD's 54.60 %. However, a more balanced energy distribution is achieved, with IMF6 ($8616.04\mu V^2$, 21.93 %) and IMF8 (5668.63 , 14.43 %) effectively capturing mid-to-high-frequency components, thus avoiding the energy concentration issues observed

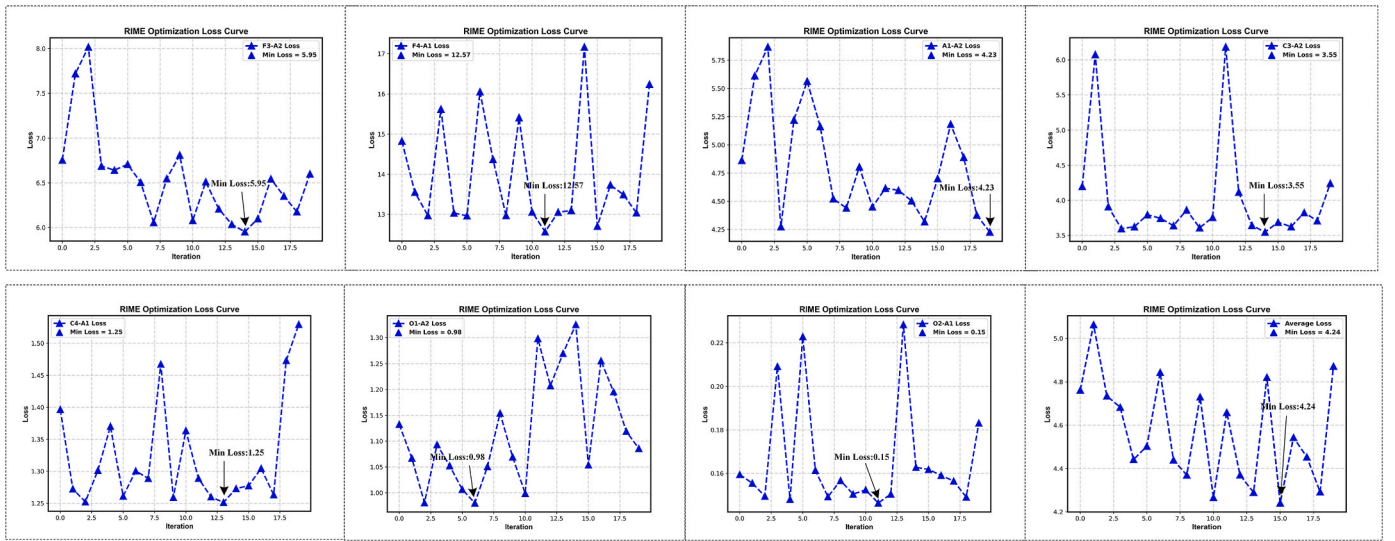


Fig. 4. Loss variation during rime optimization for EEG signal decomposition.

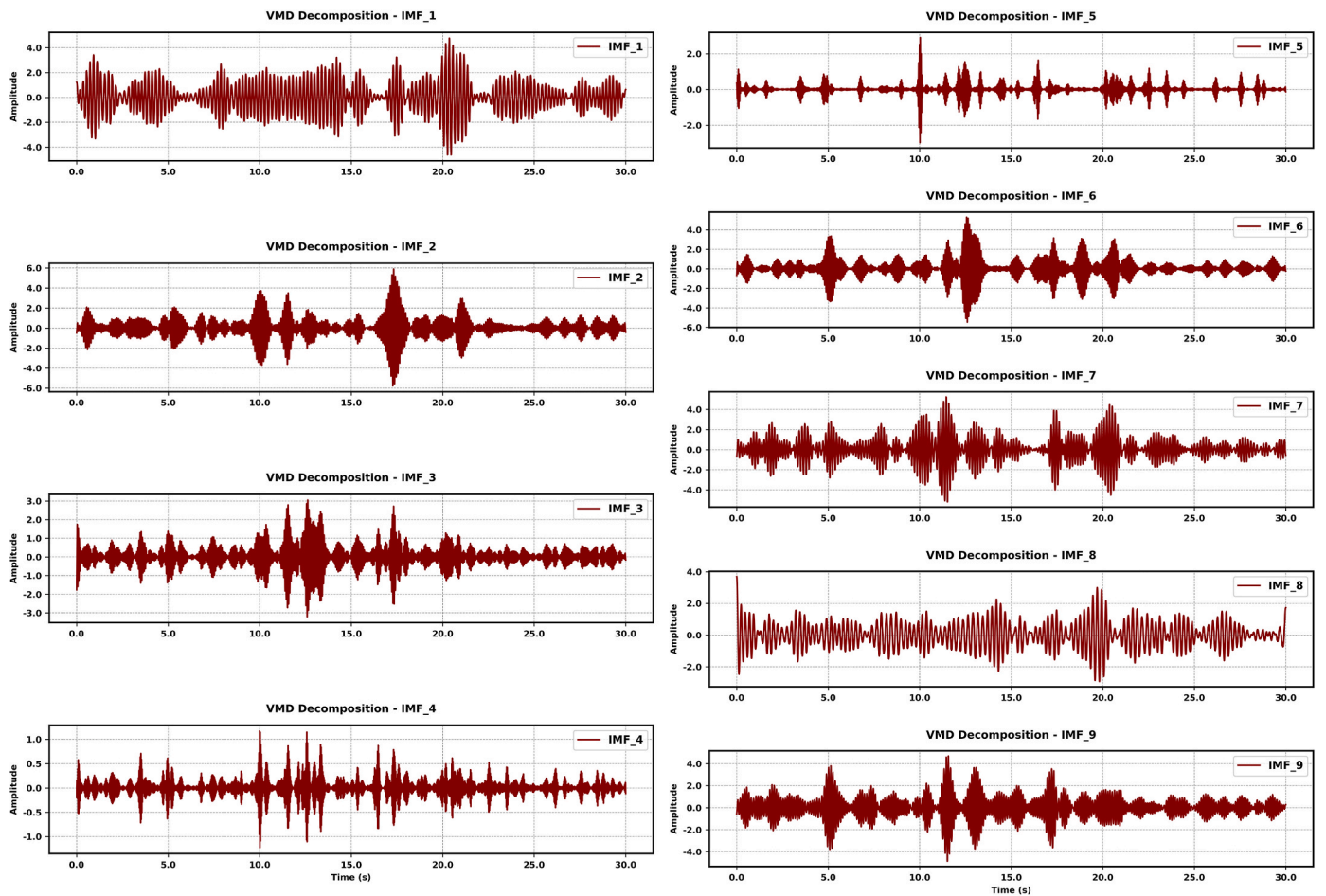


Fig. 5. IMFs from VMD of the s-EEG signal reveal frequency characteristics of sleep stages.

in GA-VMD and PSO-VMD. More importantly, relative correlations in RIME-VMD are demonstrated to be more stable (10.66 %–31.21 %), with IMF4 (10.66 %) exhibiting the highest independence and low-frequency modes (IMF1~3) showing stable correlations (23.37 %–31.21 %), indicating higher decomposition consistency compared to the significant fluctuations in PSO-VMD. This balanced distribution and

stability allow the frequency characteristics of the s-EEG signal to be comprehensively preserved.

Through the above comparison, RIME-VMD performs well in capturing low-frequency modes and achieves superior decomposition quality through optimized parameters ($K = 9$, $\alpha = 1815$), significantly outperforming traditional VMD, GA-VMD, and PSO-VMD.

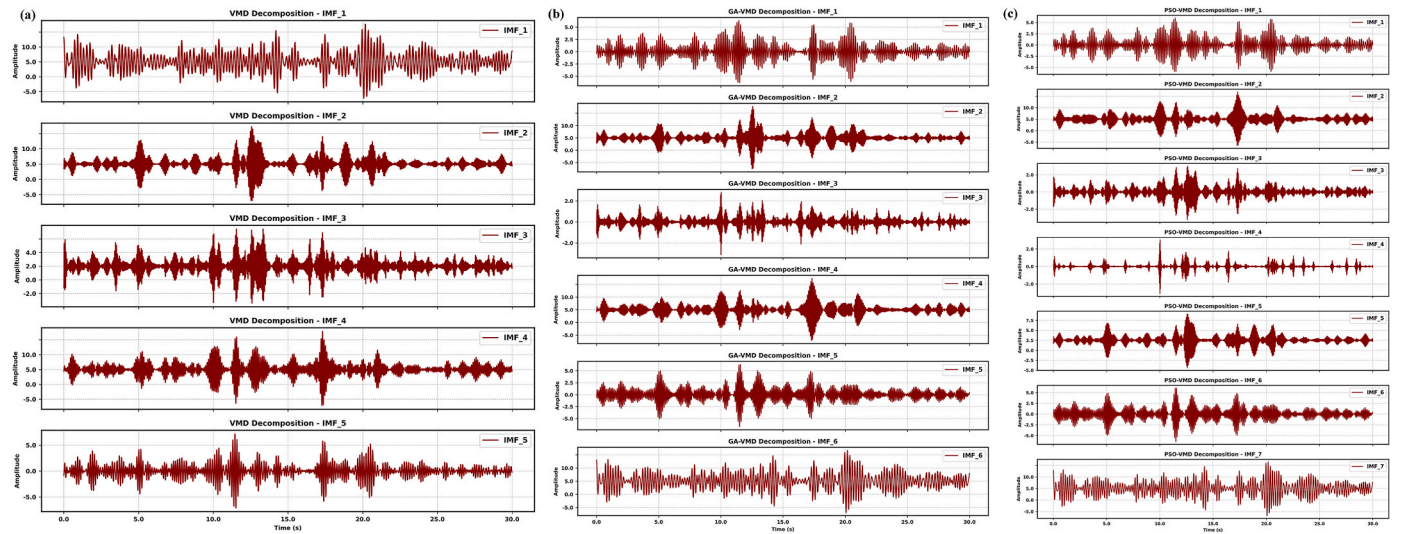


Fig. 6. Signal modal analysis using VMD and its optimized variants. (a) VMD analysis ($K = 5$, $\alpha = 2000$), yielding 5 IMFs. (b) GA-VMD analysis, yielding 6 IMFs. (c) PSO-VMD analysis, yielding 7 IMFs. All panels show a 30-s segment, with red traces representing the decomposed IMF signals.

Table 7
Energy distribution of IMFs (VMD).

IMF Index	Energy Total (uV2)	Energy Percentage (%)	Relative Correlation of IMFs (%)
1	15434.2	37.20	25.13
2	5653.40	13.63	48.27
3	1564.48	3.77	33.94
4	7085.12	17.08	49.39
5	11752.74	28.33	43.27

Table 8
Energy distribution of IMFs (GA-VMD).

IMF Index	Energy Total (uV2)	Energy Percentage (%)	Relative Correlation of IMFs (%)
1	9596.47	23.24	36.05
2	4740.83	11.48	37.79
3	991.08	2.40	24.52
4	5593.57	13.55	40.22
5	6336.92	15.35	41.22
6	14025.34	33.97	20.19

Table 9
Energy distribution of IMFs (PSO-VMD).

IMF Index	Energy Total (uV2)	Energy Percentage (%)	Relative Correlation of IMFs (%)
1	9225.69	22.31	32.14
2	5342.76	12.92	34.40
3	1758.22	4.25	30.75
4	421.58	1.02	13.18
5	4565.20	11.04	35.27
6	6228.60	15.06	36.30
7	13802.17	33.38	17.95

Table 10
Energy distribution of IMFs (RIME-VMD).

IMF Index	Energy Total (uV2)	Energy Percentage (%)	Relative Correlation of IMFs (%)
1	8454.26	21.52	31.21
2	5196.04	3.23	26.82
3	1487.14	3.79	23.37
4	412.55	1.05	10.66
5	4578.04	11.65	27.08
6	8616.04	21.93	26.80
7	4869.84	12.40	25.77
8	5668.63	14.43	28.29

mapped to distinct frequency bands via Fourier transform (slow waves: 0.5–2 Hz, θ/α waves: 4–13 Hz, β waves: 12–30 Hz). Table 11 reports the average frequency band energy distribution across the feature dimensions for a representative subset of s-EEG segments.

Table 11 reveals the energy distribution of CNN-extracted features across frequency bands, highlighting the model's ability to capture stage-specific physiological patterns. The dominance of slow waves (0.5–2 Hz) at 42.5 % in NREM stages underscores the prominence of low-frequency components, while θ/α waves (4–13 Hz) at 35.8 % and β waves (12–30 Hz) at 21.7 % align with REM and WAKE stages, respectively. This distribution validates the effectiveness of feature extraction, ensuring that the extracted features reflect the distinct frequency characteristics of each sleep stage.

Guided by the American Academy of Sleep Medicine (AASM) standards, feature selection prioritizes dimensions with high energy in stage-specific bands. Table 12 summarizes the selected feature dimensions and their correspondence to sleep stages, indicating that 65 dimensions out of 128 are selected for NREM, reflecting the prevalence of low-frequency features. In comparison, 29 and 34 dimensions are selected for REM and WAKE, respectively. This selection ensures that stage-specific characteristics are effectively captured, providing a robust foundation for subsequent dynamic weighting and classification.

4.4. Feature extraction and classification results

4.4.1. Feature extraction and selection results

Feature extraction uses a multi-layer 1D CNN on the IMFs obtained from Rime-optimized VMD, as detailed in Section 3.4.1. The CNN transforms the input IMFs shape (N, 6000, 9) into feature maps of shape (N, 1500, 128), capturing local frequency and temporal patterns. To align with stage-specific frequency characteristics, the feature maps are

Table 11
Frequency band energy distribution of CNN extracted features.

Frequency Band	Average Energy Percentage (%)	Associated Sleep Stage
Slow Waves (0.5–2 Hz)	42.5	NREM
θ/α Waves (4–13 Hz)	35.8	REM
β Waves (12–30 Hz)	21.7	WAKE

Table 12
Feature dimension selection results.

Sleep Stage	Selection Criterion	Selected Dimensions	Dominant Frequency Range (Hz)
NREM	Slow Waves	1~30, 45~60, 90~100	0.5~2 Hz
REM	θ/α Waves	31~44, 61~75	4~13 Hz
WAKE	β Waves	76~89, 101~128	12~30 Hz

4.4.2. Dynamic feature weighting results

The selected feature maps are dynamically weighted using an MHSA mechanism described in Section 3.4.2. The MHSA module, with four attention heads, assigns weights based on contextual relevance, prioritizing stage-specific patterns. Table 13 reports the average attention weights assigned to feature dimensions associated with different frequency bands across a subset of s-EEG segments.

Table 13 shows that associated with slow waves (0.5–2 Hz) receive the highest average attention weight (0.38), reflecting their prominence in NREM stages. Dimensions linked to θ/α waves (4–13 Hz) are assigned a weight of 0.25, aligning with REM characteristics, while β wave dimensions (12–30 Hz) receive a weight of 0.19, corresponding to WAKE stages. High-frequency artifacts (>30 Hz) are effectively suppressed with a minimal weight of 0.06, demonstrating MHSA's ability to prioritize stage-specific features while mitigating artifact interference. This multi-head weighting mechanism enhances the model's focus on relevant features, contributing to its high classification accuracy.

Dynamic adaptation of attention weights across time steps is illustrated in Fig. 7 for three representative s-EEG segments (1500 time steps each), where weights for slow wave (0.5–2 Hz, NREM), θ/α wave (4–13 Hz, REM), and β wave (12–30 Hz, WAKE) dimensions are plotted, revealing distinct patterns across sleep stages (NREM, REM, WAKE). In Fig. 7(a), dominated by NREM (0–900 time steps), slow waves exhibit the highest weights (averaging 0.38), consistent with Table 13, before declining during the transition to REM, where θ/α waves rise to an average of 0.25, followed by β waves increasing to 0.19 in the WAKE stage. In Fig. 7(b), dominated by REM (451–1050 time steps), θ/α waves exhibit a sustained peak (averaging 0.25), with slow waves decreasing from their initial high in the NREM stage, and β waves rising in the WAKE stage. In Fig. 7(c), dominated by WAKE (601–1499 time steps), β waves show a prominent increase (averaging 0.19), following a brief elevation of θ/α waves in the REM stage, while slow waves remain high in the initial NREM stage. These dynamic shifts, aligning with the average weighting patterns in Table 13, highlight the MHSA's capability to adaptively prioritize stage-specific features, supporting the model's effectiveness in distinguishing NREM, REM, and WAKE stages.

4.4.3. Classification results

The dynamically weighted features are processed by LSTM networks for temporal modeling, followed by classification using a fully connected layer within the RimeSleepNet framework. To ensure robustness and consistency, five-fold cross-validation was performed. Each fold was trained for 40 epochs, achieving stable convergence with an average training accuracy of 96.2 % and a training loss of 0.164 by the final epoch. Table 14 shows the training accuracy and loss curves of Fold1 to Fold5 over 40 training cycles.

Table 13
Average attention weights from MHSA.

Frequency Band	Associated Sleep Stage	Average Attention Weight
Slow Waves (0.5–2 Hz)	NREM	0.38
θ/α Waves (4–13 Hz)	REM	0.25
β Waves (12–30 Hz)	WAKE	0.19
High-Frequency Artifacts (>30 Hz)	None	0.06

Training accuracy and loss function values in Table 14 gradually increase and decrease across each fold, respectively, indicating that RimeSleepNet effectively captures s-EEG signal patterns. Specifically, the average training accuracy rises from 60.1 % at the first epoch to 92.5 % by the final epoch, while the training loss decreases from 0.674 to 0.245, reflecting stable convergence across all folds. Variability is observed in early epochs, such as a dip in Fold5 at epoch 10 (78.5 %), which reflects expected fluctuations during the training process. The consistency of the final trends across folds, with accuracy ranging from 91.8 % to 93.2 % and losses from 0.230 to 0.260, demonstrates the training process's robustness and the model's capability to learn discriminative features for sleep stage classification.

Following the training phase, the model's performance is evaluated on the test sets of the five-fold cross-validation, with each fold utilizing 80 % of the data for training and 20 % for testing. This evaluation ensures that the model's generalization ability to unseen data is robustly assessed based on the stable training results presented in Table 14. Classification performance metrics (precision, recall, and F1 scores) for NREM, REM, and WAKE stages across the five folds are presented in Table 15.

As shown in Table 15, RimeSleepNet achieved superior performance across NREM, REM, and WAKE stages. Specifically, the F1-score for NREM was 0.91 ± 0.02 (95 % CI [0.885, 0.935]), with statistical significance confirmed by two-tailed paired t-tests ($p = 0.03$). For REM, the F1-score was 0.80 ± 0.02 (95 % CI [0.775, 0.825]), and the improvement compared to baseline models was statistically significant ($p < 0.05$, Cohen's $d = 3.0$). In the WAKE category, the F1-score was 0.90 ± 0.01 (95 % CI [0.884, 0.912]), with a notably higher recall suggesting robust identification of β wave-dominated patterns. The standard deviations of precision, recall, and F1 score for NREM and REM are 0.02, 0.02, and 0.02, respectively, reflecting moderate variability across folds in the five-fold cross-validation results, which may be attributed to individual differences in s-EEG signals. In contrast, the standard deviations for WAKE are lower, indicating greater consistency in prediction, likely due to the distinct and stable characteristics of WAKE stage features.

ROC and AUC curves for the NREM, REM, and WAKE stages in both overall classification and across the five folds are presented in Fig. 8. In the overall classification, AUC values of 0.90, 0.89, and 0.91 are achieved for NREM, REM, and WAKE, respectively, indicating strong discriminative ability across all stages. Across the five folds, AUC values range from 0.88 to 0.92 for NREM, 0.87 to 0.91 for REM, and 0.89 to 0.93 for WAKE, reflecting consistent performance with minimal variability, which aligns with the F1 scores for NREM and WAKE observed in Table 15. The slightly lower AUC range for REM corresponds to its lower F1 score and recall, further highlighting the challenge of distinguishing REM due to its complex features.

The five-fold cross-validation results, including the average confusion matrix and individual confusion matrices for Fold1 to Fold5, are illustrated in Fig. 9. Average classification accuracy of 0.94, 0.87, and 0.96 are achieved for NREM, REM, and WAKE, respectively, indicating robust performance for NREM and WAKE, consistent with their F1 scores in Table 15. The lower accuracy for REM (0.87) and a misclassification rate of 0.13 reflect the difficulty in identifying this stage, corroborating the findings from Table 15. Notably, the accuracy for WAKE in Fold5 reaches 0.97, underscoring the model's effectiveness in recognizing this stage, likely due to the distinct β wave patterns.

Overall, RimeSleepNet achieves robust classification performance for NREM and WAKE stages, as evidenced by F1 scores, AUC values, and classification accuracy, demonstrating the model's effectiveness in identifying these stages. However, the consistent challenge in recognizing REM, indicated by lower F1 scores, AUC values, and higher misclassification rates, suggests that further optimization in feature extraction or data balancing may be required to improve performance for this stage.

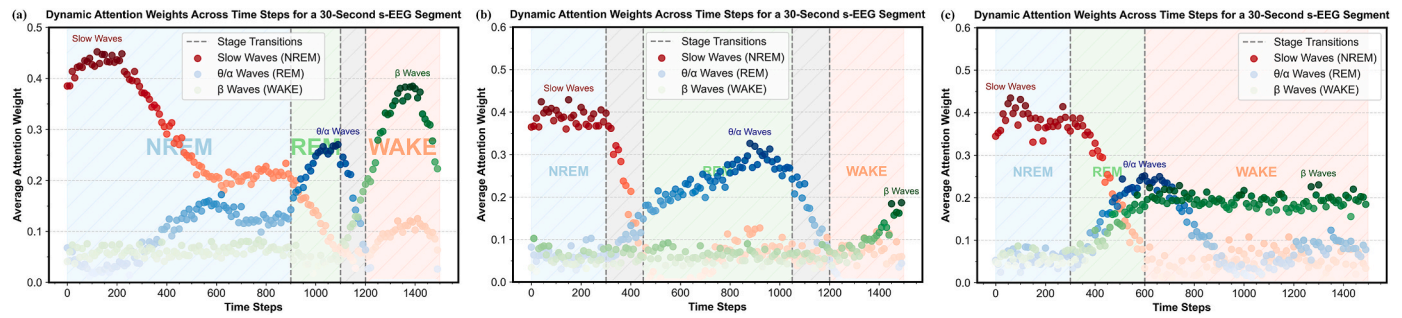


Fig. 7. Dynamic attention weights across time steps in representative s-EEG segments. (a) NREM-dominated segment. (b) REM-dominated segment. (c) WAKE-dominated segment. Each segment is annotated under AASM standards. Scatter plots show attention weights for slow wave (0.5–2 Hz, NREM), θ/α wave (4–13 Hz, REM), and β wave (12–30 Hz, WAKE) dimensions across a 30-s s-EEG segment (1500 time steps). The x-axis represents time steps, and the y-axis represents attention weight (0–0.6). Colors indicate frequency dimensions, with gradients reflecting weight magnitude; background colors denote sleep stages.

Table 14

Training accuracy and training loss of Fold1~Fold5 across 40 training epochs.

Epoch		1	10	20	30	40	Epoch		1	10	20	30	40
Fold1	Train Accuracy	60.2	82.5	88.7	91.5	92.8	Fold2	Train Accuracy	60.8	84.0	89.5	92.0	93.2
	Train Loss	0.67	0.29	0.22	0.19	0.24		Train Loss	0.67	0.27	0.21	0.19	0.23
Fold3	Train Accuracy	60.5	81.0	87.5	90.8	92.0	Fold4	Train Accuracy	59.8	83.5	88.0	91.0	92.5
	Train Loss	0.67	0.30	0.23	0.20	0.25		Train Loss	0.68	0.28	0.22	0.19	0.25
Fold5	Train Accuracy	59.0	78.5	86.0	89.5	91.8	Average	Train Accuracy	60.1	81.9	87.9	91.0	92.5
	Train Loss	0.68	0.32	0.24	0.21	0.26		Train Loss	0.67	0.29	0.22	0.20	0.25

Table 15

Performance metrics of Fold1~Fold5 for three-class classification (NREM, REM, WAKE).

Fold	NREM			REM			WAKE		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
1	0.92**	0.91**	0.92**	0.88*	0.75*	0.81	0.87*	0.94	0.90
2	0.93**	0.92**	0.92**	0.89*	0.77*	0.83	0.88*	0.95	0.91
3	0.91**	0.91**	0.91**	0.86*	0.74*	0.80	0.87*	0.93	0.90
4	0.89*	0.87*	0.88*	0.87*	0.76	0.81	0.85	0.91	0.88
5	0.92**	0.92**	0.92**	0.84	0.71	0.77	0.86*	0.93	0.89
Mean $\pm$ SD	0.91 $\pm$ 0.02	0.90 $\pm$ 0.02	0.91 $\pm$ 0.02	0.87 $\pm$ 0.02	0.75 $\pm$ 0.02	0.80 $\pm$ 0.02	0.87 $\pm$ 0.01	0.93 $\pm$ 0.01	0.90 $\pm$ 0.01

Note: Values are reported as mean $\pm$ standard deviation across five cross-validation folds. 95 % confidence intervals were calculated based on the t-distribution ($df = 4$). Pairwise model comparisons were performed using two-tailed paired t-tests for each fold's metrics, with Wilcoxon signed-rank tests used when normality assumptions were violated. For multiple comparisons, one-way repeated-measures ANOVA with Bonferroni post-hoc correction was used to adjust for multiple testing. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

4.4.4. Ablation study and cross-dataset validation

To assess the contributions of individual components in RimeSleepNet and its robustness across diverse datasets, ablation studies and cross-dataset validation are conducted. Experiments are performed on two datasets: the Chengdu People's Hospital dataset (Dataset1) and the PhysioNet Sleep-EDF dataset (Dataset2). The ablation study evaluates several variants of RimeSleepNet to quantify the impact of each component on classification performance. The variants assessed include:

- (1) LSTM only (removing RIME optimization, VMD, and MHSA weighting);
- (2) VMD-LSTM (removing RIME optimization and MHSA weighting);
- (3) RIME-VMD-LSTM (removing MHSA weighting);
- (4) The full RimeSleepNet model (RIME-VMD-MHSA-LSTM).

Classification performance metrics, including F1 scores and accuracy for NREM, REM, and WAKE stages, are analyzed on both datasets.

Fig. 10 illustrates the classification performance metrics of each variant on Dataset1 and Dataset2, clearly demonstrating the superiority of the full RimeSleepNet model over other variants. Comparing LSTM only, VMD + LSTM, and the whole model reveals that the VMD + LSTM variant improves the REM F1 score by 18.8 % and 16.4 % on Dataset1

and Dataset2, respectively, compared to LSTM only, with recall increasing by 19.9 % and 15.4 %. The error rate decreased from 0.2411 to 0.1549 (reductions of 35.8 % and 21.5 %), indicating that VMD significantly enhances feature extraction by separating s-EEG signal frequency components, particularly for REM stage identification. NREM and WAKE F1 scores also increased by 8.1 % and 16.0 % (NREM) and 9.4 % and 11.4 % (WAKE), respectively, with error rates reduced by 32.0 % to 52.5 %. Comparing RIME + VMD + LSTM and the full model shows that the full model further improves the REM F1 score by 0.6 % and 3.5 %, with precision on Dataset 2 increasing by 8.6 %. The error rate decreased from 0.1250 to 0.1231, demonstrating that RIME optimization enhances model stability through adaptive parameter tuning. At the same time, MHSA weighting further refines performance by dynamically allocating weights to temporal features, particularly improving REM precision. NREM and WAKE F1 scores increased by 1.6 % to 4.7 % (NREM) and 1.0 % to 2.3 % (WAKE), respectively, with error rates reduced by 12.5 % to 17.6 %. Overall, VMD and RIME optimization are critical for performance gains, while MHSA weighting is more effective in refining features for complex stages like REM.

Cross-dataset validation evaluates RimeSleepNet robustness on Dataset1 and Dataset2. To compare its cross-dataset performance, baseline models CNN and LSTM. Fig. 11 displays these model's training

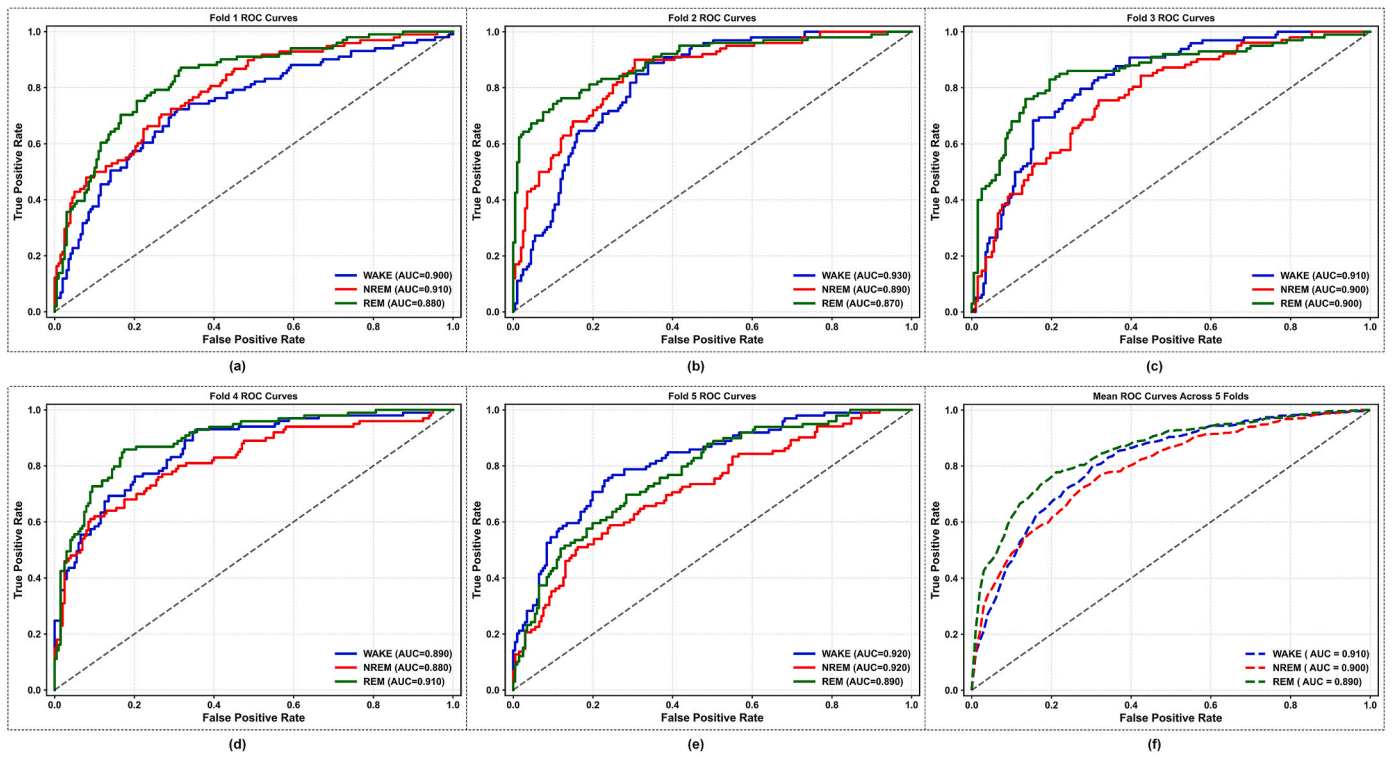


Fig. 8. ROC and AUC curves for three sleep stages (WAKE, NREM, REM) in five-fold Cross-validation (a-e) and mean ROC curve across folds (f).

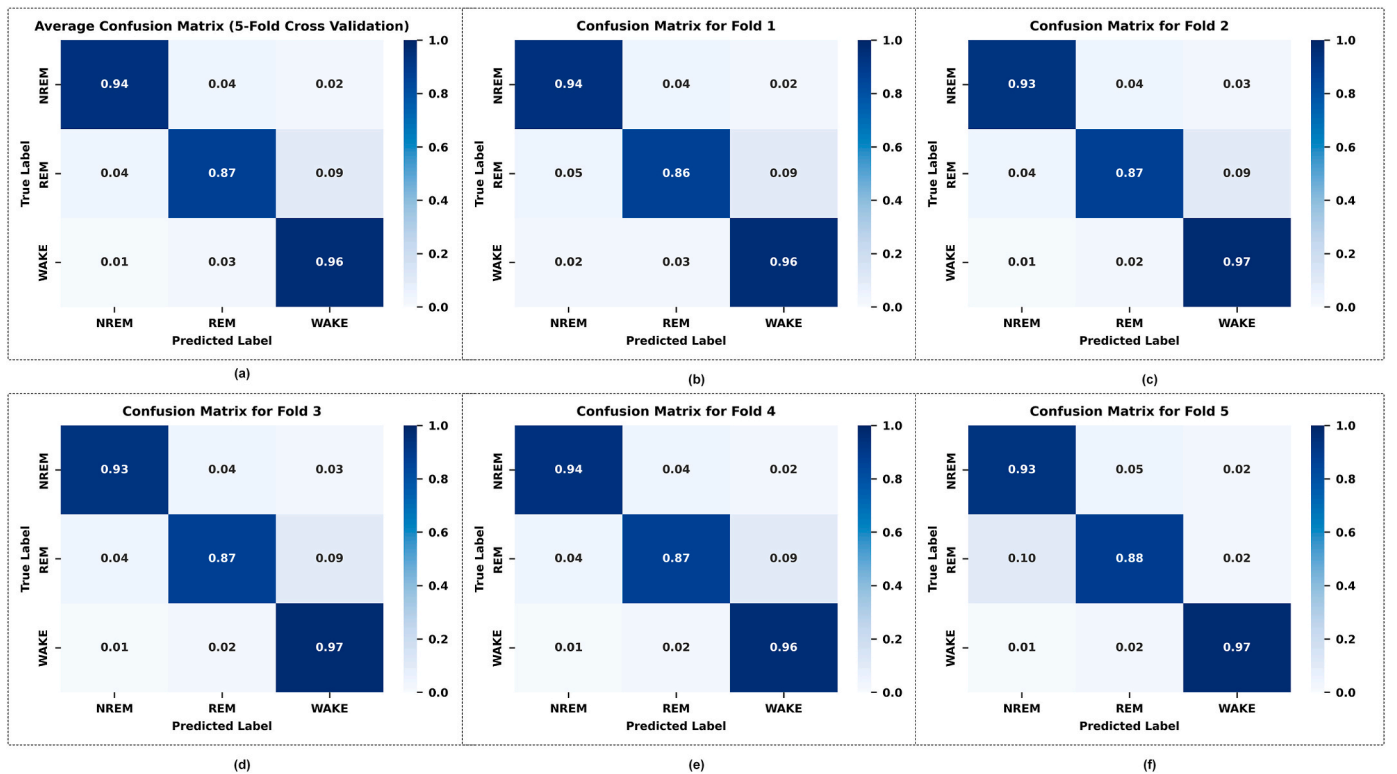


Fig. 9. Confusion matrices for sleep stage classification: average across five folds (a) and individual fold results (b ~ f).

and validation loss curves over 40 epochs on both datasets. CNN and LSTM exhibit larger cross-dataset validation loss differences (10.92 % and 36.19 %, respectively), reflecting sensitivity to data distribution, while VMD-CNN and RIME-VMD-LSTM reduce these differences to 11.92 % and 8.3 %, respectively. RimeSleepNet validation loss shows a

cross-dataset difference of only 5.08 %, demonstrating superior robustness. These findings confirm that RimeSleepNet, through the RIME-VMD-MHSA framework, effectively adapts to data distribution differences in s-EEG signal classification tasks, validating its exceptional cross-dataset generalizability.

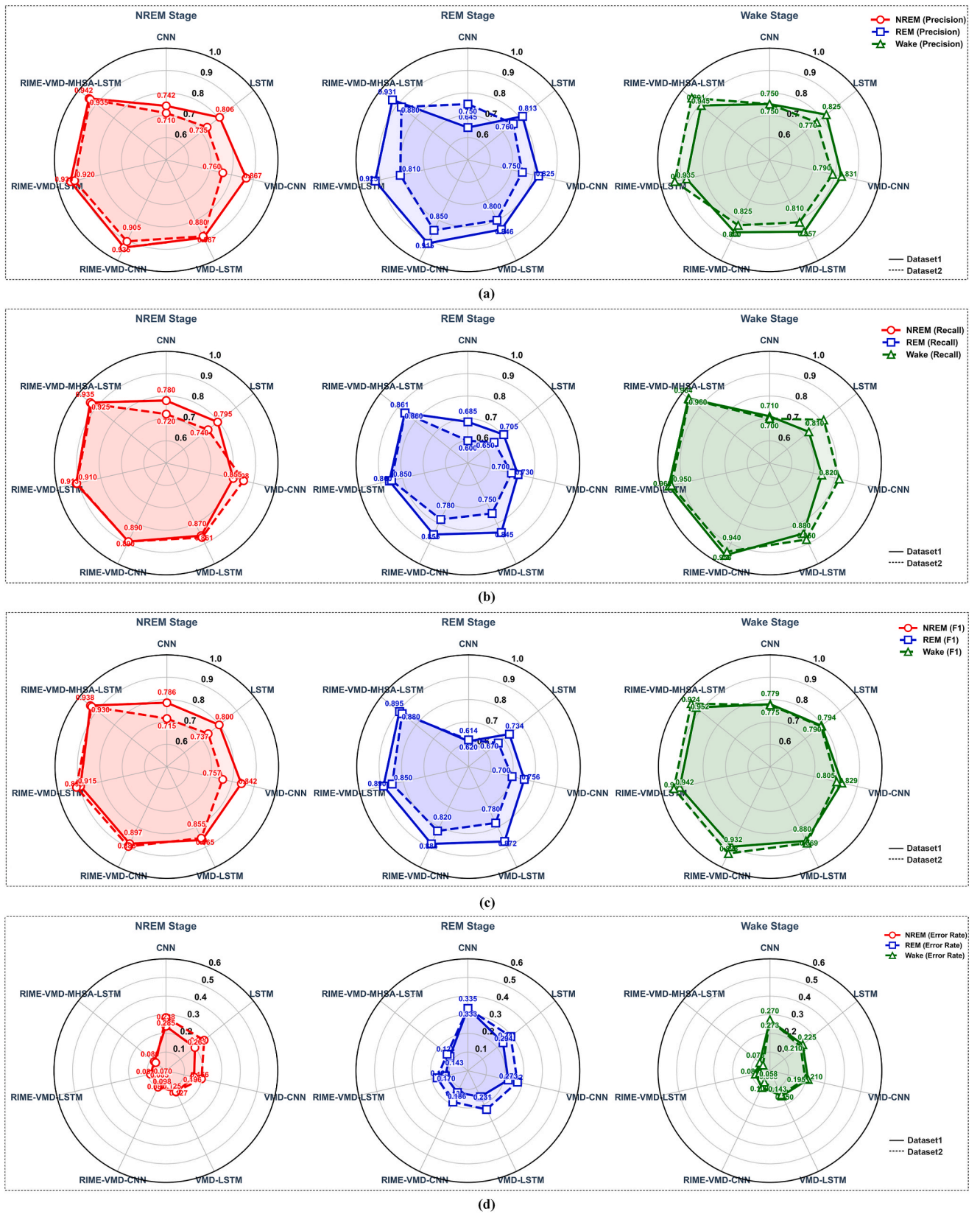


Fig. 10. Radar chart comparison of RimeSleepNet model variants' ablation performance across datasets. (a) Precision. (b) Recall. (c) F1 Score. (d) Error Rate. Radar charts display the classification performance of RimeSleepNet model variants on Dataset1 and Dataset2 across NREM, REM, and WAKE stages, totaling 12 subplots. Variants include LSTM only, CNN, VMD + LSTM, VMD-CNN, VMD-LSTM, RIME + VMD + LSTM, and RimeSleepNet. Solid and dashed lines denote Dataset1 and Dataset2, respectively, with colors indicating sleep stages, axes representing performance metrics, and angles corresponding to different variants.

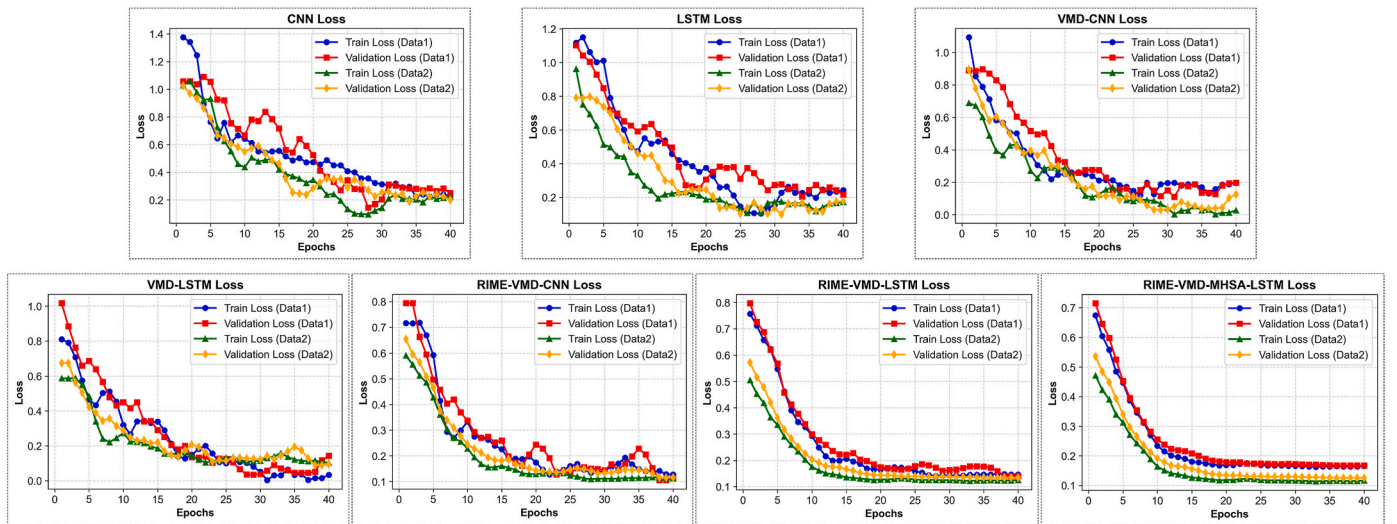


Fig. 11. Cross dataset validation loss curves of RimeSleepNet and baseline models. Loss curves display the training and validation losses of RimeSleepNet and baseline models on Dataset1 and Dataset2 across 40 epochs. Models include CNN, LSTM, VMD-CNN, VMD-LSTM, RIME-VMD-CNN, RIME-VMD-LSTM, and RimeSleepNet. Solid and dashed lines denote training and validation losses, respectively, the x-axis representing epochs and the y-axis representing loss values.

4.4.5. Comparative experiments

The performance of RimeSleepNet in s-EEG classification was validated by comparison with multiple literature methods encompassing single-modal EEG and multi-modal models, as detailed in Table 16. On the Sleep-EDF dataset, RimeSleepNet achieved an accuracy of 0.91, surpassing most single-modal models, such as AttnSleep 0.88 and SENet-HMM 0.82, and approaching EEGSNet and MML-DMS 0.94. High accuracy is attributed to RIME-optimized VMD to reduce aliasing and MHSA weighting to prioritize stage-specific features.

5. Discussion

The efficacy of RimeSleepNet in single-modal s-EEG classification was demonstrated by the highest F1-scores of 0.94 (NREM), 0.89(REM), and 0.92 (WAKE), an AUC of 0.92, and cross-dataset robustness (Cohen's $\kappa = 0.90$). The synergistic optimization of four modules drives this advance:

- (i) RIME optimization adaptively tunes VMD parameters to reduce modal aliasing;
- (ii) VMD generates artifact-resistant modes;

- (iii) CNN automatically extracts stage-specific features, and MHSA dynamically weights the extracted features to prioritize stage-specific patterns and suppress artifacts;
- (iv) LSTM models temporal dynamics to accommodate stage transitions.

CNN-RNN hybrids (such as DeepSleepNet) combine convolutional feature extraction with sequence modeling and typically report 80–85 % accuracy and 77–82 % macro-F1 on public datasets [49]. Attention-based models (such as AttnSleep) further advance temporal-context modeling and mitigate class imbalance through multi-head attention and class-sensitive training objectives [37]. Transformer-based approaches (such as SleepTransformer) provide interpretability using attention maps and enable uncertainty quantification using entropy-based decision deferral. Additionally, frameworks (such as U-Sleep) trained on large-scale datasets exhibit robust cross-dataset generalization [50,51]. Building upon these advances, RimeSleepNet integrates RIME, VMD, MHSA, and LSTM; in our evaluations, improvements in NREM and WAKE classification and reductions in fold-to-fold variability were observed.

The use of single-modal s-EEG data precludes comprehensive multimodal analysis and may limit the detection of complex sleep disorders. REM classification remains constrained by overlapping wave features. Potential confounders may persist despite exclusion of participants using medications known to alter sleep architecture or with severe psychiatric or neurological disorders; future studies should control these factors more rigorously. RimeSleepNet provides a robust tool for automated sleep analysis, facilitating insomnia diagnosis.

6. Conclusion

This paper proposes RimeSleepNet, a hybrid deep-learning model for single-modal s-EEG sleep stage classification. By integrating Rime-optimized VMD for adaptive signal analysis, MHSA for dynamic feature weighting, and LSTM for temporal dynamics modeling, RimeSleepNet effectively addresses challenges like frequency aliasing and artifact interference. Experiments on the self-collected Chengdu People's Hospital dataset and the Sleep-EDF dataset demonstrate high performance, with the highest F1 scores of 0.94, 0.89, and 0.92 for NREM, REM, and WAKE stages, respectively, and an AUC of 0.92. Cross-dataset validation confirms strong generalization (Cohen's $\kappa = 0.90$), achieving a 53 % reduction in validation loss compared to LSTM

Table 16
Classification performance comparison of RimeSleepNet with other methods.

Model	Dataset	Modality	Classification Accuracy
AttnSleep [37]	Sleep-EDF-20, Sleep-EDF-78, SHHS	EEG	0.88
TCNN-CRF [38]	Sleep-EDF-2013, Sleep-EDF-2018	EEG	0.85
SENet-HMM [39]	sleep-EDF	EEG	0.82
EOGNET [40]	MASS SS3	EOG	0.85
SleepContextNet [41]	Sleep-EDF-201	EEG	0.85
EEGSNet [42]	Sleep-EDFX-8	EEG	0.94
SAN [43]	MASS	EEG	0.87
MaskSleepNet [44]	Sleep-EDFX	EEG	0.84
TSA-Net [45]	Sleep-EDF-20	EEG	0.87
MML-DMS [46]	PhysioNet CAP Sleep	EEG	0.94
ST-GATv2 [47]	MASS SS3	EEG\EOG	0.90
1D-CNN [48]	Sleep-EDF	EEG\EOG	0.94
RimeSleepNet	Sleep-EDF	EEG	0.91

baselines. Ablation studies highlight the essential roles of RIME optimization and MHSA in boosting classification accuracy. Despite limitations in REM classification due to feature overlap, RimeSleepNet, through optimized feature extraction and dynamic weighting, delivers a robust solution for automated sleep analysis, supporting sleep disorder diagnosis and personalized monitoring. This study focuses on single-modal s-EEG, which offers high temporal resolution but lacks complementary physiological channels such as EOG and EMG; this limitation may hinder the capture of eye-movement and muscle-tone features that are informative for certain stages. In addition, the current analysis is based on a single-centre clinical cohort combined with a public dataset, and the overall sample size remains moderate. Future work will enlarge the cohort through multi-centre collaborations and integrate multi-modal inputs (such as EOG, EMG) to further strengthen generalizability and the detection of complex sleep disorders.

CRediT authorship contribution statement

Xiaolin Wang: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Xiaowei Li:** Writing – review & editing, Writing – original draft, Methodology. **Jing Li:** Formal analysis, Data curation. **Yunfa Fu:** Writing – review & editing, Writing – original draft, Supervision, Methodology. **Dan Zhang:** Writing – review & editing. **Yan Peng:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] Zia Jasmine K, et al. Risk factors for abdominal pain-related disorders of gut–brain interaction in adults and children: a systematic review. *Gastroenterology* 2022;163(4):995–1023. <https://doi.org/10.1053/j.gastro.2022.06.028>.
- [2] Wang Shanze, et al. The obesity-related dietary pattern is associated with higher risk of sleep disorders: a cross-sectional study from NHANES. *Nutrients* 2022;14(19):3987. <https://doi.org/10.3390/nu14193987>.
- [3] Sletten Tracey L, et al. The role of circadian phase in sleep and performance during antarctic winter expeditions. *J Pineal Res* 2022;73(2):e12817. <https://doi.org/10.1111/jpi.12817>.
- [4] Brink-Kjaer Andreas, et al. Age estimation from sleep studies using deep learning predicts life expectancy. *npj Digit Med* 2022;5(1):103. <https://doi.org/10.1038/s41746-022-00630-9>.
- [5] Lopresti Adrian L. The effects of psychological and environmental stress on micronutrient concentrations in the body: a review of the evidence. *Adv Nutr* 2020;11(1):103–12. <https://doi.org/10.1093/advances/nmz082>.
- [6] Yang Yuzhe, et al. Artificial intelligence-enabled detection and assessment of Parkinson's disease using nocturnal breathing signals. *Nat Med* 2022;28(10):2207–15. <https://doi.org/10.1038/s41591-022-01932-x>.
- [7] Hassan Fatima, Hussain Syed Fawad, Qaisar Saeed Mian. Fusion of multivariate EEG signals for schizophrenia detection using CNN and machine learning techniques. *Inf Fusion* 2023;92:466–78. <https://doi.org/10.1016/j.inffus.2022.12.019>.
- [8] Ranjan Rakesh, Sahana Bikash Chandra, Bhandari Ashish Kumar. Cardiac artifact noise removal from sleep EEG signals using hybrid denoising model. In: *IEEE transactions on instrumentation and measurement*; 2022. p. 1–10. <https://doi.org/10.1109/TIM.2022.3198441>.
- [9] Dash Shaswati, et al. Time–frequency domain machine learning for detection of epilepsy using wearable EEG sensor signals recorded during physical activities. *Biomed Signal Process Control* 2025;100:107041. <https://doi.org/10.1016/j.bspc.2024.107041>.
- [10] Schmierer Thomas, Li Tianning, Li Yan. A novel empirical wavelet SODP and spectral entropy based index for assessing the depth of anaesthesia. *Health Inf Sci Syst* 2022;10(1):10. <https://doi.org/10.1007/s13755-022-00178-8>.
- [11] Piho Laura, Tjahjadi Tardi. A mutual information based adaptive windowing of informative EEG for emotion recognition. *IEEE Trans Affect Comput* 2018;11(4):722–35. <https://doi.org/10.1109/TAFFC.2018.2840973>.
- [12] Du Xiaobing, et al. An efficient LSTM network for emotion recognition from multichannel EEG signals. *IEEE Trans Affect Comput* 2020;13(3):1528–40. <https://doi.org/10.1109/TAFFC.2020.3013711>.
- [13] He Mingyuan, et al. TBSTSleepNet: three-Branch spectro-temporal bidirectional LSTM based attention model for EEG sleep staging. *Biomed Signal Process Control* 2024;97:106695. <https://doi.org/10.1016/j.bspc.2024.106695>.
- [14] Wang Yao, et al. BI-Directional long short-term memory for automatic detection of sleep apnea events based on single channel EEG signal. *Comput Biol Med* 2022;142:105211. <https://doi.org/10.1016/j.combiomed.2022.105211>.
- [15] Lee Ji Hyun, et al. Develop a deep learning model for sleep stage prediction in obstructive sleep apnea cohort using 60 GHz frequency-modulated continuous-wave radar. *J Sleep Res* 2024;33(1):e14050. <https://doi.org/10.1111/jsr.14050>.
- [16] Chen Xueyan, et al. Sleep staging by bidirectional long short-term memory convolution neural network. *Future Gener Comput Syst* 2020;109:188–96. <https://doi.org/10.1016/j.future.2020.03.019>.
- [17] Wang Eileen, Irena Koprinska, Jeffries Bryn. Sleep apnea prediction using deep learning. In: *IEEE journal of biomedical and health informatics*; 2023. p. 5644–54. <https://doi.org/10.1109/JBHI.2023.3305980>.
- [18] Brandmayr Georg, et al. Relational local electroencephalography representations for sleep scoring. *Neural Netw* 2022;154:310–22. <https://doi.org/10.1016/j.neunet.2022.07.020>.
- [19] Efe Enes, Ozsen Seral. CoSleepNet: automated sleep staging using a hybrid CNN-LSTM network on imbalanced EEG-EOG datasets. *Biomed Signal Process Control* 2023;80:104299. <https://doi.org/10.1016/j.bspc.2022.104299>.
- [20] Mahmud Tanvir, et al. Sleep apnea detection from variational mode decomposed EEG signal using a hybrid CNN-BiLSTM. *IEEE Access* 2021;9:102355–67. <https://doi.org/10.1109/ACCESS.2021.3097090>.
- [21] Kumar Gulshan, Chander Subhash, Almadhor Ahmad. An intelligent epilepsy seizure detection system using adaptive mode decomposition of EEG signals. *Phys Eng Sci Med* 2022;45(1):261–72. <https://doi.org/10.1007/s13246-022-01111-9>.
- [22] Sharma Hemant, Sharma KK. Sleep apnea detection from ECG using variational mode decomposition. *Biomed Physics Eng Express* 2020;6(1):015026. <https://doi.org/10.1088/2057-1976/ab68e9>.
- [23] Bhalerao Shailesh Vitthalrao, Pachori Ram Bilas. Sparse spectrum based swarm decomposition for robust nonstationary signal analysis with application to sleep apnea detection from EEG. *Biomed Signal Process Control* 2022;77:103792. <https://doi.org/10.1016/j.bspc.2021.103792>.
- [24] Khare Smith K, Rajendra Acharya U. An explainable and interpretable model for attention deficit hyperactivity disorder in children using EEG signals. *Comput Biol Med* 2023;155:106676. <https://doi.org/10.1016/j.combiomed.2023.106676>.
- [25] Alyasser Zaid Abdi Alkareem, et al. EEG signals denoising using optimal wavelet transform hybridized with efficient metaheuristic methods. *IEEE Access* 2019;8:10584–605. <https://doi.org/10.1109/ACCESS.2019.2962658>.
- [26] Subasi Abdulhamit, Kevric Jasmin, Abdullah Canbaz M. Epileptic seizure detection using hybrid machine learning methods. *Neural Comput Appl* 2019;31:317–25. <https://doi.org/10.1007/s00521-017-3003-y>.
- [27] Xiong Xin, et al. Research on ocular artifacts removal from single-channel electroencephalogram signals in obstructive sleep apnea patients based on support vector machine, improved variational mode decomposition, and second-order blind identification. *Sensors* 2024;24(5):1642. <https://doi.org/10.3390/s24051642>.
- [28] Liu Wenhui, et al. A wind speed forecasting model based on rime optimization based VMD and multi-headed self-attention-LSTM. *Energy* 2024;294:130726. <https://doi.org/10.1016/j.energy.2024.130726>.
- [29] Cai Chenhao, Zhang Leyao, Zhou Jianguo. DMPR: a novel wind speed forecasting model based on optimized decomposition, multi-objective feature selection, and patch-based RNN. *Energy* 2024;310:133277. <https://doi.org/10.1016/j.energy.2024.133277>.
- [30] Thomas CW, Blanco-Duque C, Bréant BJ, Goodwin GM, Sharp T, Bannerman DM, Vyazovskiy VV. Psilocin acutely alters sleep-wake architecture and cortical brain activity in laboratory mice. *Transl Psychiatry* 2022;12:77. <https://doi.org/10.1038/s41398-022-01846-9>.
- [31] Sikka A, Jamalabadi H, Krylova M, et al. Investigating the temporal dynamics of electroencephalogram (EEG) microstates using recurrent neural networks. *Hum Brain Mapp* 2020;41:2334–46. <https://doi.org/10.1002/hbm.24949>.
- [32] Su Hang, et al. RIME: a physics-based optimization. *Neurocomputing* 2023;532:183–214. <https://doi.org/10.1016/j.neucom.2023.02.010>.
- [33] Wang Yanxue, et al. Research on variational mode decomposition and its application in detecting rub-impact fault of the rotor system. *Mech Syst Signal Process* 2015;60:243–51. <https://doi.org/10.1016/j.ymssp.2015.02.020>.
- [34] Iber C, Ancoli-Israel S, Chesson AL, Quan SF. The AASM manual for the scoring of sleep and associated events: rules, terminology and technical specifications. first ed. American Academy of Sleep Medicine; 2007. <https://api.semanticscholar.org/CorpusID:78939623>.

- [35] Goldberger AL, Amaral LA, Glass L, Hausdorff JM, Ivanov PC, Mark RG, et al. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *Circulation* 2000;101:e215–20. <https://doi.org/10.1161/01.cir.101.23.e215>.
- [36] Kemp B, Zwinderman AH, Tuk B, Kamphuisen HAC, Obery J. Analysis of a sleep-dependent neuronal feedback loop: the slow-wave microcontinuity of the EEG. *IEEE (Inst Electr Electron Eng) Trans Biomed Eng* 2000;47:1185–94. <https://doi.org/10.1109/10.867928>.
- [37] Eldele E, Ragab M, Chen Z, Wu C, Tan MK, Tian X, et al. An attention-based deep learning approach for sleep stage classification with single-channel EEG. In: *IEEE trans. Neural syst. Rehabil. Eng.*; 2021. p. 809–18. <https://doi.org/10.1109/TNSRE.2021.3076234>. 29.
- [38] Khalili Elham, Asl Babak Mohammadzadeh. Automatic sleep stage classification using temporal convolutional neural network and new data augmentation technique from raw single-channel EEG. *Comput Methods Progr Biomed* 2021;204: 106063. <https://doi.org/10.1016/j.cmpb.2021.106063>.
- [39] Yang Bufang, Zhu Xilin, Liu Yitian, et al. A single-channel EEG based automatic sleep stage classification method leveraging deep one-dimensional convolutional neural network and hidden markov model. *Biomed Signal Process Control* 2021; 68:102581. <https://doi.org/10.1016/j.bspc.2021.102581>.
- [40] Fan Jiahao, Sun Chenglu, Meng Long, et al. EOGNET: a novel deep learning model for sleep stage classification based on single-channel EOG signal. *Front Neurosci* 2021;15:573194. <https://doi.org/10.3389/fnins.2021.573194>.
- [41] Zhao Caihong, Li Jinbao, Guo Yahong. SleepContextNet: a temporal context network for automatic sleep staging based single-channel EEG. *Comput Methods Progr Biomed* 2022;220:106806. <https://doi.org/10.1016/j.cmpb.2022.106806>.
- [42] Shi J, Zhao Y, Wang C, Zeng H, Dai G. EEGSNet: a novel EEG cognitive recognition model using spiking neural network. *Biomed Signal Process Control* 2025;105: 107610. <https://doi.org/10.1016/j.bspc.2025.107610>.
- [43] Zhou W, Zhu H, Shen N, Chen H. A lightweight segmented attention network for sleep staging by fusing local characteristics and adjacent information. In: *IEEE trans. Neural syst. Rehabil. Eng.*; 2022. <https://doi.org/10.1109/TNSRE.2022.3220372>. 99.
- [44] Zhu H, Liu Y, Zhang X, Wang J, Chen L, et al. MaskSleepNet: a cross-modality adaptation neural network for heterogeneous signals processing in sleep staging. *IEEE J Biomed Health Inform* 2023;27:2353–64. <https://doi.org/10.1109/JBHI.2023.3253728>.
- [45] Fu G, Zhou Y, Gong P, Wang P, Shao W, Zhang D, et al. A temporal-spectral fused and attention-based deep model for automatic sleep staging. *IEEE Trans Neural Syst Rehabil Eng* 2023;31:1008–18. <https://doi.org/10.1109/TNSRE.2023.3238852>.
- [46] Cheng YH, Lech M, Wilkinson RH. Simultaneous sleep stage and sleep disorder detection from multimodal sensors using deep learning. *Sensors* 2023;23:3468. <https://doi.org/10.3390/s23073468>.
- [47] Hu Y, Shi W, Yeh CH. Spatiotemporal convolution sleep network based on graph attention mechanism with automatic feature extraction. *Comput Methods Progr Biomed* 2024;244:107930. <https://doi.org/10.1016/j.cmpb.2023.107930>.
- [48] Yildirim O, Baloglu UB, Acharya UR. A deep learning model for automated sleep stages classification using PSG signals. *Int J Environ Res Publ Health* 2019;599. <https://doi.org/10.3390/ijerph16040599>.
- [49] Supratak A, Dong H, Wu C, Guo Y. DeepSleepNet: a model for automatic sleep stage scoring based on raw single-channel EEG. *IEEE Trans Neural Syst Rehabil Eng* 2017;25:1998–2008. <https://doi.org/10.1109/TNSRE.2017.2721116>.
- [50] Phan H, Mikkelsen K, Chen OY, Koch P, Mertins A, De Vos M. SleepTransformer: automatic sleep staging with interpretability and uncertainty quantification. *IEEE (Inst Electr Electron Eng) Trans Biomed Eng* 2022;69:2456–67. <https://doi.org/10.1109/TBME.2022.3147187>.
- [51] Perslev M, Darkner S, Kempfner L, et al. U-Sleep: resilient high-frequency sleep staging. *npj Digit Med* 2021;4:72. <https://doi.org/10.1038/s41746-021-00440-5>.